Software Test Plan (STP)

Project	IT Infrastructure & Asset Management System
Version	1.0
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1. Introduction

1.1 Purpose

This test plan defines the testing scope, approach, environment, schedule, responsibilities, risks and acceptance criteria for ITAMS v1.0.

1.2 Scope

Testing covers authentication and RBAC, asset registration and search, allocation/return, software license and warranty records, maintenance scheduling/history, alerts, dashboards/reports, import/export, audit trail, performance, usability and security. Direct endpoint discovery, procurement/accounting integrations and external identity/email provider internals are excluded; their interfaces may be stubbed.

1.3 References and Definitions

References: ITAMS SRS v1.0 and Software Architecture and Design Specification v1.0. Definitions: STP—Software Test Plan; RTM—Requirements Traceability Matrix; RBAC—Role-Based Access Control; UAT—User Acceptance Testing.

2. Test Items

- Authentication and role/permission enforcement
- Asset inventory CRUD, validation, search and filters
- Allocation, return, reassignment and retirement
- Software license entitlement and expiry records
- Warranty tracking
- Maintenance schedule and service history
- Alert generation, acknowledgement and resolution
- Dashboard, reports, CSV/PDF export and CSV import
- Audit events and log handling
- Performance, browser usability, security and backup/restore procedures

3. Features to be Tested

Feature	SRS Traceability	Test Cases
Authentication/RBAC	ITAM-F-001, ITAM-F-002; ITAM-SR-002	TC-AUTH-01, TC-SEC-01
Asset creation and validation	ITAM-F-003 to ITAM-F-005	TC-ASSET-01 to TC-ASSET-03
Search/filter	ITAM-F-006	TC-SEARCH-01
Allocation and return	ITAM-F-007, ITAM-F-008	TC-ALLOC-01, TC-ALLOC-02
License and entitlement compliance	ITAM-F-009, ITAM-F-010	TC-LIC-01, TC-COMP-01
Warranty and maintenance	ITAM-F-011 to ITAM-F-013	TC-WAR-01, TC-MAINT-01, TC-MAINT-02

Alerts	ITAM-F-014 to ITAM-F-016	TC-ALERT-01 to TC-ALERT-03
Dashboard and reports	ITAM-F-017 to ITAM-F-019	TC-DASH-01, TC-REPORT-01, TC-REPORT-02
Audit and lifecycle	ITAM-F-020, ITAM-F-021	TC-AUDIT-01, TC-LIFE-01
CSV import	ITAM-F-022	TC-IMPORT-01
Performance	ITAM-NF-001	TC-PERF-01
Security and confidentiality	ITAM-NF-003, ITAM-NF-006; ITAM-SR-001 to ITAM-SR-006	TC-SEC-02 to TC-SEC-06
Accessibility/browser compatibility	ITAM-NF-005	TC-UX-01
Backup and recovery	ITAM-NF-007	TC-REC-01

4. Features Not to be Tested

- Internal behaviour of external SMTP providers, identity providers or database engines beyond the integration contract.
- Physical hardware inventory scanning and automatic endpoint discovery unless implemented in the project.
- Procurement, accounting, HR and enterprise CMDB integrations outside the approved baseline.
- Production-scale availability claims; a student test environment can validate mechanisms but cannot prove long-term production uptime.

5. Test Approach / Strategy

5.1 Test Levels

- Unit testing: validate business rules, date thresholds, input validation and permission helpers.
- Integration testing: validate service-to-database operations, transaction consistency and optional notification adapters.
- System testing: execute end-to-end workflows through the user interface/API.
- Acceptance testing: demonstrate representative asset-manager, employee and auditor tasks to the project guide/team.

5.2 Test Types

- Functional and negative testing; regression testing after fixes.
- Security testing: authentication, authorization, injection payloads, session handling, secret/log inspection and TLS configuration where deployed.
- Performance testing: response-time measurement for common searches and list/report endpoints against a defined test dataset.
- Usability and compatibility testing: browser checks, keyboard navigation, clear validation messages and responsive layout.
- Data integrity testing: uniqueness, referential integrity, history retention and allocation state transitions.
- Recovery testing: backup and restoration of representative records, if supported by the deployment.

5.3 Entry and Exit Criteria

Entry criteria: approved SRS and design available; stable build deployed; test accounts/roles and synthetic asset data prepared; database initialized; test environment and defect tracker accessible.

Exit criteria: 100% of planned test cases executed or formally deferred; all critical defects closed; all high-priority requirements pass or have documented approval for waiver; RTM updated; no unresolved data-loss or privilege-escalation issue.

5.4 Representative Test Cases

Test ID	Scenario	Steps / Setup	Expected Result	Execution Status
TC-AUTH-01	Valid login	Submit valid test credentials	User reaches dashboard; session is established.	Pass/Fail

TC-SEC-01	RBAC denial	Login as employee and attempt admin-only asset edit via UI/API	Request is denied server-side; no record changes.	Pass/Fail
TC-ASSET-01	Create asset	Submit valid required asset fields	Asset ID is generated and record is retrievable.	Pass/Fail
TC-ASSET-03	Duplicate tag	Create a second asset with an existing asset tag	System rejects or clearly flags duplicate; no silent duplicate.	Pass/Fail
TC-ALLOC-01	Allocate available asset	Allocate available asset to valid employee	Current holder and allocation history update consistently.	Pass/Fail
TC-ALLOC-02	Return asset	Return currently allocated asset	Status becomes available/returned and prior history remains.	Pass/Fail
TC-LIC-01	License expiry data	Create license with entitlement and expiry date	Saved license details are displayed to authorized role.	Pass/Fail
TC-COMP-01	Excess entitlement	Set recorded usage above entitlement quantity	Compliance view flags over-allocation.	Pass/Fail
TC-MAINT-01	Schedule maintenance	Create a due maintenance item for an asset	Item appears with due date, priority and status.	Pass/Fail
TC-ALERT-01	Expiry alert	Set license expiry within configured threshold and run alert job	One matching alert is generated; repeat run does not duplicate it.	Pass/Fail
TC-ALERT-02	Overdue maintenance	Create overdue maintenance record and run alert job	Alert appears with asset and due-date details.	Pass/Fail
TC-REPORT-01	Report accuracy	Generate department inventory report using known test data	Rows and totals match the source records and filter.	Pass/Fail
TC-AUDIT-01	Audit lifecycle action	Edit asset and inspect audit view	Actor, timestamp, action and target record are recorded.	Pass/Fail
TC-IMPORT-01	CSV validation	Import file containing valid and invalid rows	Valid rows import; invalid rows are identified with actionable errors.	Pass/Fail
TC-PERF-01	Search response time	Run repeated asset searches against 10,000 synthetic assets	95th percentile is at or below 3 seconds in the documented test environment.	Pass/Fail
TC-SEC-02	Injection handling	Submit SQL/script-like strings in searchable/form fields	Inputs are safely handled; no unauthorized query execution or script execution.	Pass/Fail
TC-SEC-03	Secret logging	Perform login and license operations, then inspect logs	No passwords, session tokens or full license secrets appear.	Pass/Fail
TC-UX-01	Keyboard and validation usability	Navigate forms without mouse and submit invalid data	Focus order and labels are usable; errors are clear and associated with fields.	Pass/Fail
TC-REC-01	Backup/restore	Back up test database, restore to test instance and compare records	Representative assets, assignments and history are restored consistently.	Pass/Fail

6. Security Validation

- Verify TLS/HTTPS is enabled in the deployed environment and weak configurations are not accepted.
- Verify role checks are performed by the server for read, create, update, retire, export and audit endpoints.
- Try malformed inputs, oversized CSV files and injection-like strings using synthetic data only.
- Inspect application logs and error responses for credentials, session tokens, license secrets, stack traces or unnecessary personal data.
- Verify session expiry/logout and login failure handling where implemented.
- Confirm audit events exist for privileged changes and cannot be modified by ordinary users.

7. Test Environment

Area	Planned Setup
Hardware	Developer/test workstation; no dedicated physical asset hardware required.
Software	ITAMS test build, supported browser(s), relational database and synthetic test dataset.
Tools	Postman or equivalent API client; browser developer tools; framework test runner; spreadsheet for execution evidence; Git issue tracker or defect log.
Test data	Synthetic employees, departments, hardware assets, software licenses, warranties, maintenance events and role accounts. Do not use real credentials or confidential company data.
Configuration	Document build/version, database seed version, browser version, test date and test dataset size with each run.

8. Test Schedule

Proposed relative schedule; replace with actual team dates after implementation milestones are agreed.

Milestone	Target Window	Deliverable
Test planning and RTM review	Week 1	Approved test plan and mapped test cases
Environment and synthetic data setup	Week 1	Repeatable test environment
Unit and integration testing	Weeks 2–3	Test logs and defect records
System, security and regression testing	Weeks 3–4	Execution evidence and resolved defects
Performance and usability testing	Week 4	Measurements and usability findings
Acceptance demonstration and sign-off	Week 5	Final test summary and updated RTM

9. Test Deliverables

- Software Test Plan
- Test case catalogue and test data
- Unit/integration/system test results
- Defect log with severity, steps, owner and status
- Performance measurements and environment details
- Security and usability checklist results
- Updated Requirements Traceability Matrix
- Final Test Summary Report and acceptance sign-off

10. Roles and Responsibilities

Role	Name	Responsibility
Project / QA Coordinator	SRI RAM V	Coordinate plan, test coverage, schedule and final report.
Test Engineer	Sharvesh	Prepare and execute functional test cases; record defects and retest fixes.
Developer / Test Support	Yashwanth	Support test setup, unit/integration tests, debugging and defect fixes.
Developer / Documentation Support	Shreyan Ganesh	Maintain traceability, test evidence, regression checks and documentation.

11. Risks and Mitigation

Risk	Mitigation
Build or features delayed	Prioritize high-risk/high-priority flows and test incrementally against stable builds.
Incomplete or poor-quality seed data	Use a controlled synthetic dataset and document expected record counts.
Email integration unavailable	Test notification generation through in-app alerts or a stub adapter.
Environment instability	Keep a reproducible setup guide and database seed/backup.
Requirement changes	Update SRS IDs, RTM and impacted test cases under version control.
Limited time for security testing	Prioritize authorization, injection handling, secret logging and transport security checks.

12. Assumptions and Dependencies

- The SRS and design specification are the baseline for expected behaviour.
- The team will provide a stable build, synthetic data and test accounts for each role.
- External notification services are optional and may be mocked.
- Expiry thresholds and performance measurements will be documented before execution.
- Test results apply to the tested build and environment; they do not certify production readiness by themselves.

13. Suspension and Resumption Criteria

Suspend testing if the test environment is unavailable for more than four hours, the build blocks more than 30% of planned cases, data corruption makes results unreliable, or a severe security issue requires containment.

Resume after the blocking defect is resolved, environment/data integrity is restored, a smoke test passes and the team records the new build/version.

14. Test Case Management, Traceability and Metrics

Each test case shall have a unique ID, requirement mapping, preconditions, steps, expected result, actual result, status, tester and execution date. Defects shall include severity, reproduction steps, evidence and retest status.

- Metrics: tests planned/executed/passed/failed/blocked; requirement coverage; defect count by severity; defect aging; regression pass rate; measured p95 response time.
- Reporting: brief execution update after each test cycle and a final summary stating scope, environment, results, unresolved risks and acceptance recommendation.

15. Approvals

Role	Name	Signature / Date
Project Guide	To be assigned	
Team Representative	SRI RAM V	
QA/Test Coordinator	SRI RAM V	